Mehul Dadhich

AI / ML Engineer | Computer Vision • LLM Agents • RAG • Model Inference
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Summary

AI/ML Engineer who builds AI systems end to end and ships them to production. At Infrax.ai, built **ASTRA**, a 53-tool LLM operations agent on a self-hosted Qwen3-4B, and the real-time computer-vision module of a traffic management system on live CCTV. Independently built an offline detection + VLM accident pipeline, a hybrid-RAG assistant with a QLoRA fine-tune, and a voice AI interviewer. Measures every change: held-out benchmarks, latency percentiles, tool-calling accuracy and automated tests.

Experience

Infrax.ai **Jan 2026 – Jul 2026**
AI Engineer Intern *Remote*

ASTRA – AI Operations Agent for the Cygnus Video Management System | *Qwen3-4B, llama.cpp, Node.js/Fastify, FastAPI, BM25, bge-small, Vitest, pytest*

- Designed and built **ASTRA**, an LLM agent inside Cygnus that lets operators query and control cameras, recordings, events and incidents in plain language. It runs on a self-hosted **Qwen3-4B (Q4_K_M) on llama.cpp on CPU**, with no third-party AI API.
- Implemented a **tool-calling agent loop** over **53 tools**. Each tool calls the platform's real API routes **with the operator's own token**, so permissions and site scope always apply. Writes (PTZ moves, renames, number-plate tracking) run only after a **human-in-the-loop confirmation card**.
- Built a **docs RAG service** in FastAPI: **BM25 + bge-small** embeddings fused with **reciprocal rank fusion**. A rule-based fast path answers common queries from live data without calling the model, and keeps working when the LLM is down.
- Benchmarked tool calling on a **42-case eval set**: **85.7% pass**, **88.1% correct tool**, **92.9% valid JSON**. Cut agent prompts ~**70%** (4.3k → 1.3k tokens) and used KV-cache reuse to bring a cold prompt down from **62.9 s to 0.88 s** when warm. Covered by **250+ Vitest/pytest tests**.

TMCS – Computer-Vision Module of an Advanced Traffic Management System | *YOLOv8, YOLO11, OpenCV, MediaMTX, FFmpeg, FastAPI, PostgreSQL*

- Built **YOLOv8/YOLO11** detection on **live RTSP CCTV** to identify road users and flag stopped vehicles, accidents, fire, smoke and fallen objects, with continuous video ingestion through OpenCV, MediaMTX and FFmpeg.
- Converted detections into incident events served through **FastAPI + PostgreSQL** to operator dashboards and real-time alerts. Tuned the pipeline for 24/7 operation (frame latency, stream drop-outs, inference throughput).

Projects

RoadGuard – Offline Two-Tier Accident Detection | *PyTorch, YOLOv8, ByteTrack, Qwen3-VL, TensorRT, OpenCV*

- Architected a **two-tier real-time pipeline**: two **YOLOv8s** detectors (one fine-tuned on Indian highway cameras, **mAP50 0.77**) → **ByteTrack** → a kinematic rules engine (km/h speed drops, new overlaps, vanishing vehicles, multi-signal fusion). It raises alerts instantly and sends uncertain events to a local **Qwen3-VL-2B VLM** on a priority-queue worker thread.
- Cut VLM verification latency **11× (5.5 s → 0.47 s p50)** by reformulating a JSON prompt into a constrained yes/no query. Diagnosed 4-bit quantization as the cause of false confirmations: switching to bf16 took normal-traffic rejection from **0/11 to 11/11**.
- Ran both detectors and the VLM in **4.26 GB** on a 6 GB laptop GPU, at **25.2 ms/frame (p50)** for Tier 1. Evaluated on **262 held-out benchmark clips** and a **6.5-hour shadow run on 5 live site cameras**. Hardened for 24/7 use: RTSP reconnect with backoff, health checks, log rotation, **30 pytest tests**.

Arc – Self-Hosted RAG Assistant with QLoRA Fine-Tuning | *FastAPI, llama.cpp, Llama 3.2 3B, PEFT, BM25, Firebase*Live

- Built a grounded **RAG assistant** with **hybrid retrieval** (BM25 + bge-small embeddings, project-aware boosting, relevance floor) and a 3,000-token prompt budget. It streams answers from **Llama 3.2 3B (4-bit GGUF)** via llama.cpp and is deployed on an Oracle Cloud ARM VM with no third-party AI API.
- **QLoRA fine-tuned** Qwen2.5-1.5B (4-bit NF4, rank-16 LoRA, 18.5M trainable params) on 745 grounded examples. Fact recall on 105 held-out questions rose **71%** → **90%** (document questions) and **46%** → **76%** (reasoning). Rejected a later Qwen3-4B fine-tune that scored below its base model.
- Added per-visitor conversation memory in Firestore (anonymous auth, isolated per user) and **human-in-the-loop page actions**, backed by pytest suites for retrieval, prompts, memory and API auth.

Rehearse – Voice AI Mock Interviewer | *FastAPI, Groq gpt-oss-120b, Whisper, React, Redis, Vercel* Live | GitHub

- Built a voice-first LLM interviewer whose **server-side state machine** controls the conversation (questions, at most 2 follow-ups per question, wrap-up). Every model reply is **validated against a JSON Schema** and retried if invalid or truncated.
- Voice I/O uses the Web Speech API with a server-side **Whisper large-v3-turbo** fallback. An **LLM-as-judge** scores the transcript against a 5-dimension rubric. Deployed serverless on Vercel with Redis-backed sessions and per-IP rate limits.

AI Task Manager – LLM Agent for Task Planning | *LangGraph, LangChain, Gemini, Node.js, Firebase, Socket.IO* GitHub

- Built an **agentic LangGraph workflow** that turns plain-language instructions into structured tasks (priority, category, due date), coordinating state and tool execution across steps. **Google Gemini** handles classification and natural-language date extraction, with a keyword-rule fallback so the app keeps working when the LLM is unavailable. Node.js/Express backend with Firebase Auth/Firestore and **Socket.IO** real-time alarms.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL
ML & Deep Learning: PyTorch, Hugging Face Transformers, PEFT (LoRA/QLoRA), bitsandbytes, NumPy, fine-tuning & evaluation
Computer Vision: YOLOv8/YOLO11 (Ultralytics), OpenCV, object detection & tracking (ByteTrack), OCR/ANPR, VLMs (Qwen3-VL)
GenAI & LLMs: AI agents, tool calling, LangGraph, LangChain, RAG, hybrid search (BM25 + dense, RRF), embeddings, ColPali, OKF (Open Knowledge Format), structured outputs, LLM-as-judge
Inference & Serving: llama.cpp, GGUF quantization, TensorRT (INT8), FP16/bf16, CUDA, KV-cache reuse, latency optimisation, Prometheus
Backend & Data: FastAPI, Node.js (Fastify/Express), REST, WebSockets, PostgreSQL, Redis, Firebase/Firestore, React, Next.js
Video & DevOps: RTSP, FFmpeg, MediaMTX, Docker Compose, Nginx, Git, Linux, Vercel, Oracle Cloud, pytest, Vitest

Education

VIT-AP University **2022 – 2026**
B.Tech. in Computer Science Engineering – Artificial Intelligence & Machine Learning *CGPA: 7.67/10*

DAV Public School, Ashok Vihar, Delhi **2020 – 2021**
Class XII – CBSE *86%*

DAV Public School, Ashok Vihar, Delhi **2018 – 2019**
Class X – CBSE *90.2%*

Certifications

- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional** – Oracle
- Hedera Certified Developer Associate (HCDA)
- Generative AI and ChatGPT – BlackBucks Education